

Module 5: Parsing

Basic concepts: Context-Free Grammar and English Grammar Rules; Top-down and bottom-up parsing, treebank; Syntactic parsing; working of CKY parsing. Case Study: Application of Parsing in a text processing task.

1. Explain the fundamental concepts of syntactic and semantic parsing in NLP.
2. Discuss the differences between constituency and dependency parsing. Provide examples to illustrate.
3. Explain the CYK parsing algorithm and its key components. How does CYK parsing handle context-free grammars?
4. Provide a step-by-step walkthrough of the CYK parsing process with a concrete example.
5. Analyse the time and space complexity of the CYK parsing algorithm. How does the complexity scale with the length and complexity of the input sentence?
6. Discuss potential optimizations or modifications to the CYK algorithm to improve its efficiency.
7. Discuss how CYK parsing handles ambiguity in context-free grammars. Provide examples to illustrate cases of ambiguity and how CYK resolves them.
8. Compare CYK with other parsing algorithms in terms of its ability to handle ambiguous grammars.
9. Explain how probabilistic parsing models, like the probabilistic context-free grammar (PCFG), enhance parsing accuracy. Provide examples.
10. Explore recent advancements in deep learning-based parsing techniques. How do neural network architectures, such as recurrent neural networks (RNNs) and transformers, contribute to parsing accuracy?
11. Discuss the challenges and solutions in parsing languages with free word order, such as languages with rich morphological structures.
12. Describe the role of semantic parsing in extracting meaning from natural language text. Provide examples of applications where semantic parsing is crucial.
13. Discuss the challenges of mapping natural language to formal meaning representations and propose potential solutions.
14. Explain the importance of domain-specific parsing in specialized NLP applications (e.g., biomedical text, legal documents). How can parsing models be adapted to handle domain-specific challenges?
15. Discuss the role of domain-specific lexicons and ontologies in improving parsing accuracy.